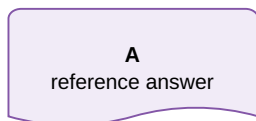
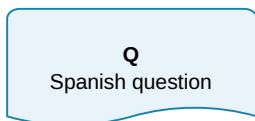
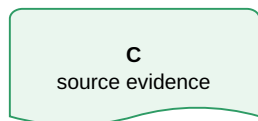


Grounded QLoRA fine-tuning

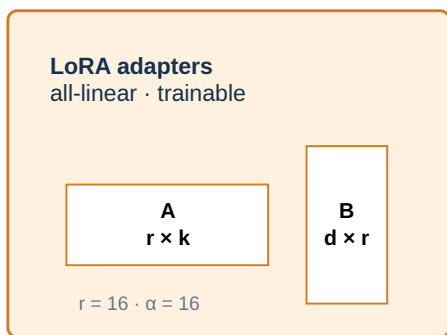
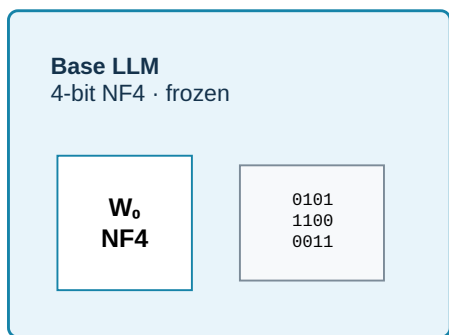
Evidence-conditioned supervision updates adapters, not the 4-bit backbone

01 GROUNDED TRAINING SAMPLE



$$s_i = [c_i \ q_i \ a_i]$$

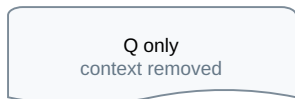
02 QLoRA OPTIMIZATION



$$\min_{A,B} - \sum_t \log p_{W_0, A, B}(a_{i,t} \mid c_i, q_i, a_{i,1:t-1})$$

03 Grounded QLoRA checkpoint base model reference + compact PEFT adapter

04 CLOSED-BOOK SFT ABLATION



$$s_i^{cB} = [q_i \ a_i]$$

Only A and B receive optimizer updates; the quantized backbone remains frozen.